F71 BF2

May 2016 solutions

Q1

(i) $(P-300)\,\ddot{a}^{(12)}_{55:\overline{10}|}{}^{7.618} = 50,000\,A^{1}_{55:\overline{10}|}{}^{.56877}$

$= £4,033.07$

(ii) $P\,\ddot{a}_{55:\overline{10}|}{}^{8.219} = 50,000\,A_{55:\overline{10}|}{}^{.65588}$

$P = £4,160.36$

$V(0) = 0$

$V(2) = 50,000\,A_{57:\overline{8}|}{}^{0.73704} - 4160.36\,\ddot{a}_{57:\overline{8}|}{}^{6.838}$

$= £8,401.96$

$V(10) = 50,000$ [or zero]

need to explain that this is after the benefit payment.

Q 2

(i) $\frac{dV}{dt} = \overset{0.05}{\delta} V(t) - t + \mu_{x+t} V(t)$

(ii) $V(n) = 0$

(iii) $V(n-h) = V(n) - h \left.\frac{dV}{dt}\right|_{t=n}$

$= \underline{\underline{nh}}$

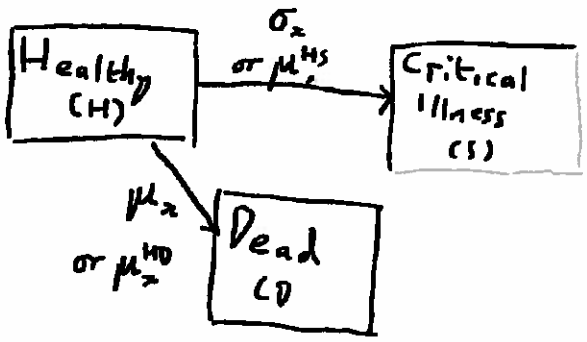
$V(n-2h) = V(n-h) - h \left.\frac{dV}{dt}\right|_{t=n-h}$

$= nh - h(\delta_{nh-(n-h)} + (\mu_{x+n-h})_{nh})$

$= 2nh - h^2(\delta_{n+1} + n\mu_{x+n-h})$

Q3

(i)



There should be no arrow from S to D, because this is not part of the benefit structure.

(ii) $EPV = \int_0^{\infty} e^{-\delta t} \, {}_t p_x^{HH} \, (150{,}000 \, \sigma_{x+t} + 80{,}000 \mu_{x+t}) dt$

Q4

(i) $\bar{a}_{72} \approx \ddot{a}^{(m)}_{72} - \frac{1}{2}$ (10.711)

$= 10.211$

$EPV = 12,000 \times 10.211 = £122,532$

(ii) $\bar{a}_{72|68} \approx \ddot{a}_{72|68}$

$= \ddot{a}^{r}_{68}$ (13.723) $- \ddot{a}_{72:68}$

$\ddot{a}_{72:68} \approx \frac{3}{5} \ddot{a}_{70:66}$ (10.368) $+ \frac{2}{5} \ddot{a}_{75:71}$ (8.238)

$= 9.516$

$\therefore EPV = 9,000 \times (13.723 - 9.516)$

$= £37,863$

(iii) Mortality is projected to the year 2020.

Q5

(i) $e^{-0.052} \times 100,000 = \underline{\underline{94,933}} = l_{41}$

(ii) $l_{42} = 94,933 \times e^{-.043} = 90,937$

$\therefore d_{41} = 94,933 - 90,937$

$= 3,996$

$\text{Deaths} = \frac{.03}{.043} \times 3996$

$= \underline{\underline{279}}$

Question 6

Death rate	Withdrawal rate
0.01	0.15

x	50	51	52
lx	1000	840	705.6

Unit fund for customer

Year	1	2	3
Value of units just before start of year	0	2686.97	6787.03
Allocation percentage	66	96	126
Allocation	2640.00	3840.00	5040.00
Bid/offer spread	132.00	192.00	252.00
Value of units just after allocation	2508.00	6334.97	11575.03
Value of units just before year end	2708.64	6841.77	12501.04
Management charge at year end	21.67	54.73	100.01
Value of units at year end	2686.97	6787.03	12401.03

Cash-flows to life office

Year	1	2	3
Unallocated premium	1360.00	160.00	-1040.00
Bid/offer spread	132.00	192.00	252.00
Expenses	-400	-100	-100
Commission	-£160	-£80	-£80
Cash-flow at start of year	932.00	172.00	-968.00
Management charge at year end	21.67	54.73	100.01
Extra death benefit	£93.13	£52.13	£0.00
Profit vector	£888.50	£179.76	-£897.03

Commission %	4%	2%	2%

Parameters

Premium	£4,000
Bid/offer spread	5%
Growth rate	8%
Management charge	0.80%
Interest for life office	3%
Minimum death benefit	£12,000
Risk discount rate	10%

Profit figures

Year	1	2	3	Total
Profit vector	£888.50	£179.76	-897.03	
Probability in force at start of year	1	0.84	0.71	
Profit signature	£888.50	£151.00	-632.95	
EPV of profit	807.73	124.80	-475.54	456.98

Reserves needed

Year	1	2	3
Reserve needed after profit vector, per policy in force at start of year	450.01	731.56	0
Reserve needed at end of year, just before profit vector	0.00	551.80	897.03
Reserve needed at start of year	0.00	535.72	870.90

Cash-flows allowing for reserves

Year	1	2	3	
Profit vector	438.49	0.00	0.00	
Profit signature	438.49	0	0	
EPV of profit	398.63	0.00	0.00	398.63

Q7

(i) $CDR = \frac{560}{45,000} = 0.012444$

(ii) $CDR = \frac{140,000}{11m} = 0.012727$

(iii) Expected deaths. $\frac{10K \times 30K}{4m} = 75$

$+ 90$

$+ 100$

265

$\therefore$ Ratio $= \frac{560}{265} = 2.1132$

(iv) Miners' mortality is heavier, but all workers have a lower CDR.

This appears to be because they are much younger than the population.